

The Limit of a Function (1.4)



Today's Target

"I can calculate a limiting value of a function both graphically and algebraically.
I can apply what I have learned in familiar and unfamiliar settings."

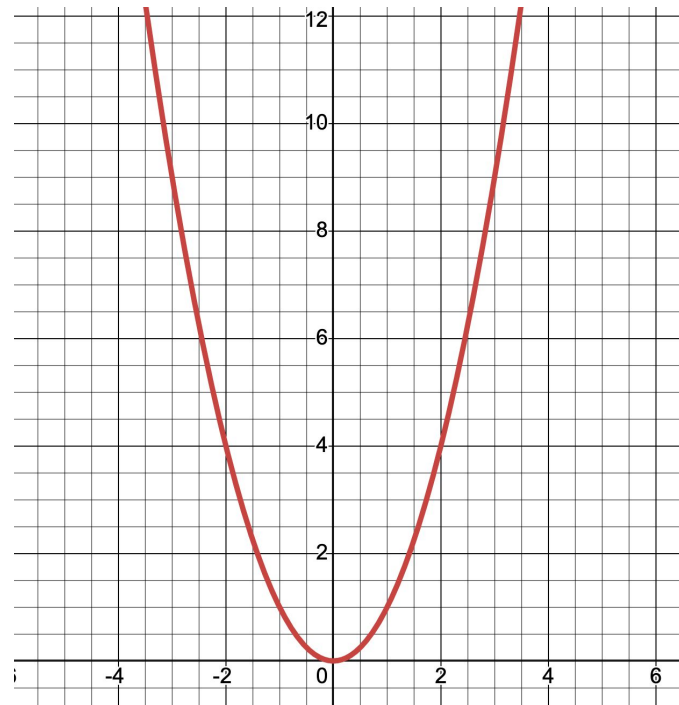
A1. Rates of Change



What is the limit, given $f(x)=x^2$

$$\lim_{x \rightarrow 4} f(x)$$

$$\lim_{x \rightarrow 2^-} f(x)$$



What is the limit, given $f(x)=x^2$

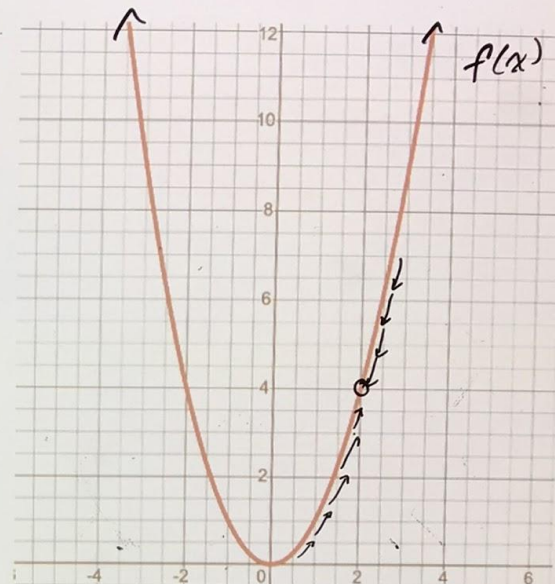
$$\lim_{x \rightarrow 2} f(x) = 4$$

$$\lim_{x \rightarrow 2^-} f(x) = 4$$

approaching
from the
left (or
from below)

$$\lim_{x \rightarrow 2^+} f(x) = 4$$

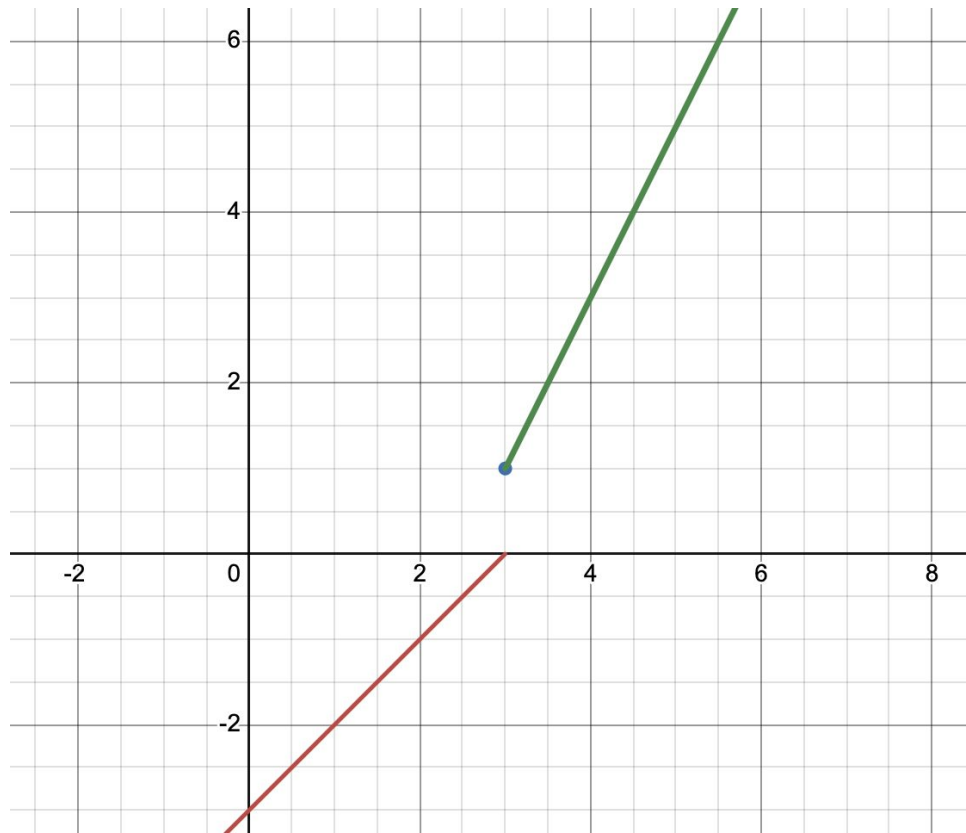
right-hand
limit
→ approach
from above



Given the following function, what is the limit...

As x approaches 2

As x approaches 3?



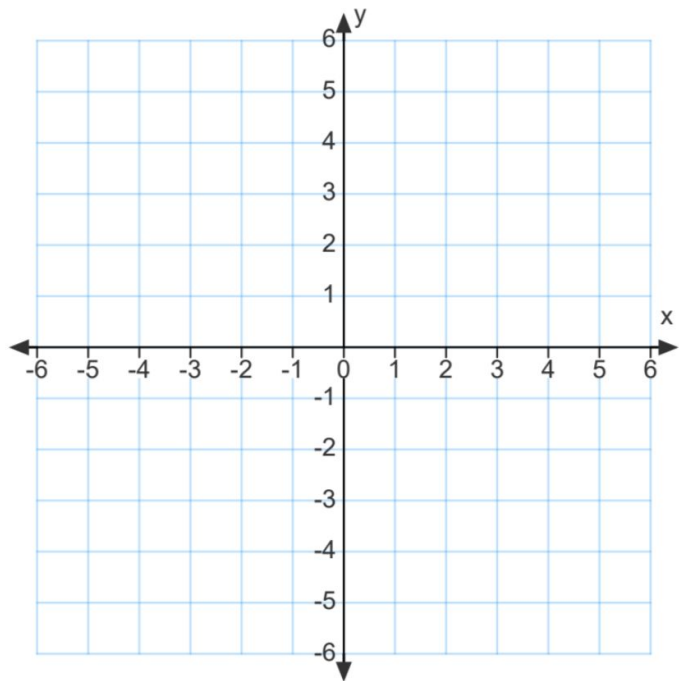
Example

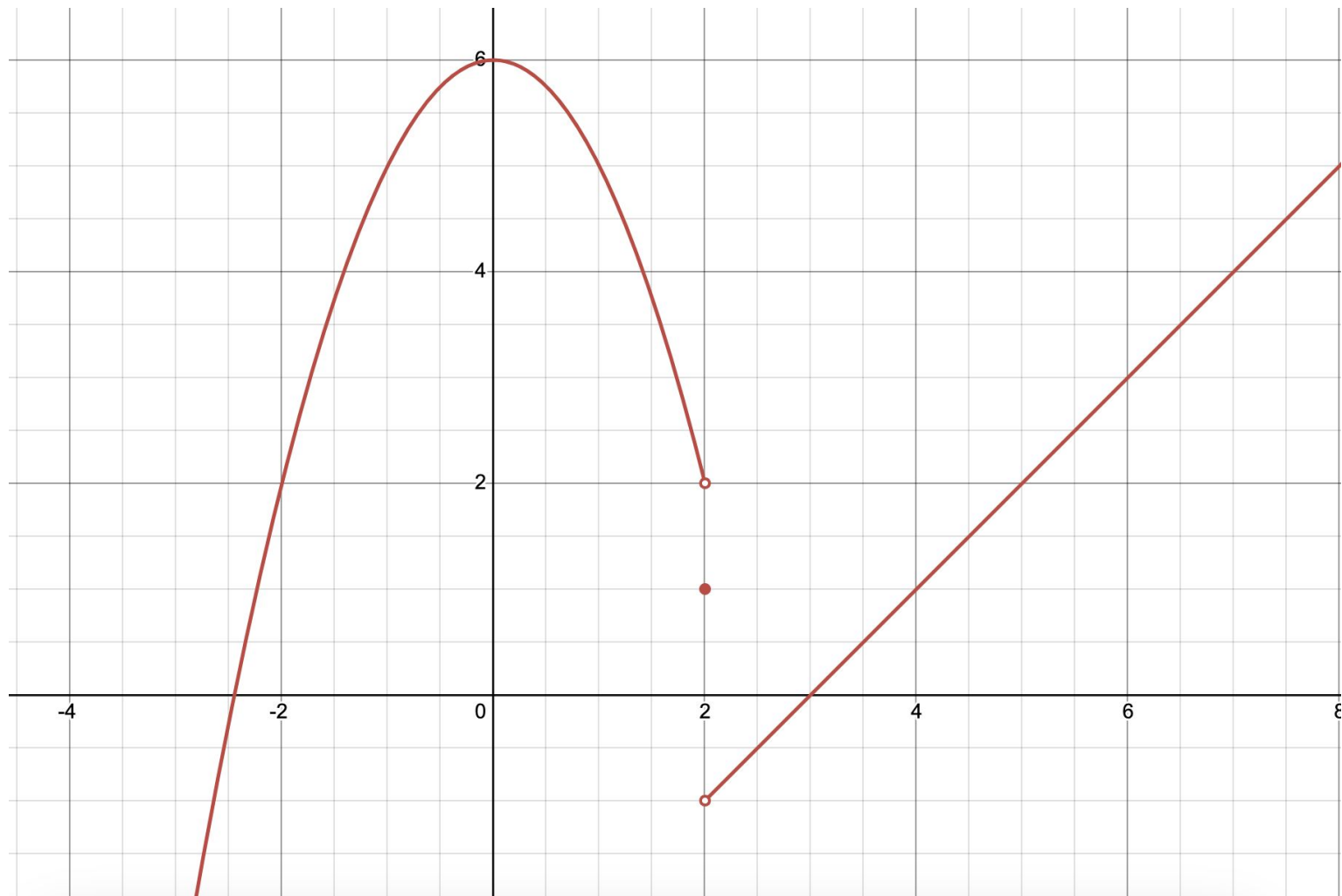
a) Graph

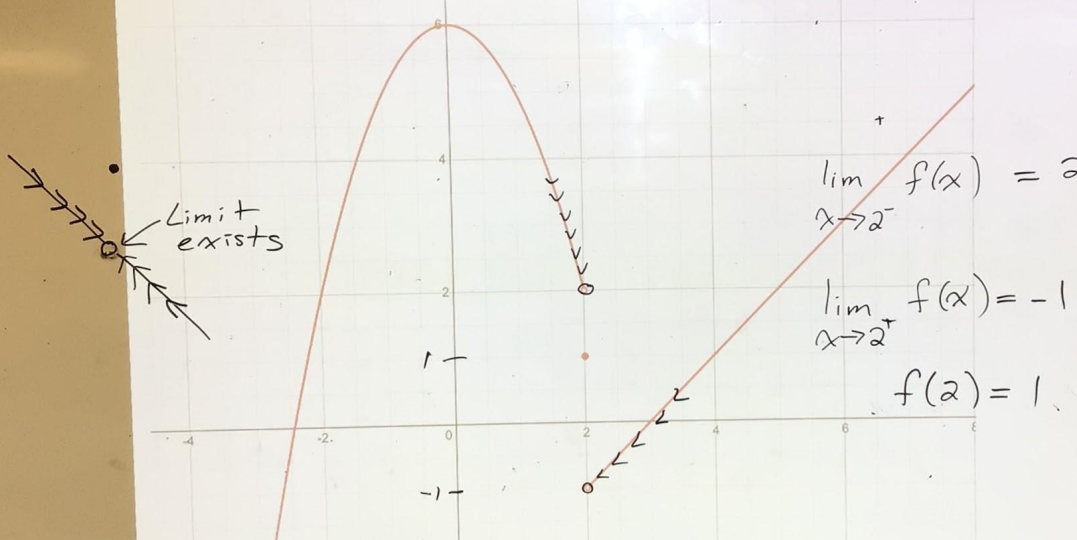
$$f(x) = \begin{cases} 6 - x^2, & x < 2 \\ 1, & x = 2 \\ x - 3, & x > 2 \end{cases}$$

b) Evaluate, if it exists.

$$\lim_{x \rightarrow 2} f(x)$$







$$\lim_{x \rightarrow 2} f(x) = \text{DNE}$$

$$f(x) = \begin{cases} 6 - x^2, & x < 2 \\ 1, & x = 2 \\ x - 3, & x > 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} f(x)$$

$$= \lim_{x \rightarrow 2^-} 6 - x^2$$

$$= 6 - 4$$

$$= 2$$

$$\lim_{x \rightarrow 2^+} f(x)$$

$$= \lim_{x \rightarrow 2^+} x - 3$$

$$= 2 - 3$$

$$= -1$$

$$\therefore \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

$$\therefore \lim_{x \rightarrow 2} f(x) \text{ DNE}$$

Limit of a Function

(simply stated¹)

Limits and Their Existence

We say that the number L is the limit of a function $y = f(x)$ as x approaches the value a , written as $\lim_{x \rightarrow a} f(x) = L$, if $\lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$. Otherwise, $\lim_{x \rightarrow a} f(x)$ does not exist.

Note: L must be a real number.

Putting the "fun" in function....

Worksheet (Level 1, 2 and 3 difficulty) +

Page 37...#

4f	5	8bc	10df	11c	12c	14	15
<i>1</i>	<i>3</i>	<i>1</i>	<i>1,3</i>	<i>3</i>	<i>3</i>	<i>4+</i>	<i>4+</i>

Level

Review it on your own - [CEMC](#)